

Immigrant Petition for Alien Workers
Employment-Based Second Preference (EB-2)
National Interest Waiver (NIW)

Pursuant to Section 203(b)(2) of the Immigration and Nationality Act
8 U.S.C. § 1153(b)(2) and 8 C.F.R. § 204.5(k)

Petitioner and Beneficiary: Vivek Kumar Mishra
Classification Sought: Employment-Based Immigration, Second Preference
National Interest Waiver (EB-2 NIW)
Sec. 203(b)(2) INA [8 U.S.C. 1153(b)(2)]

To whom it may concern,

This letter is respectfully submitted in support of the petition of Mr. Vivek Kumar Mishra for classification as a qualified immigrant under the second preference employment immigration category pursuant to section 203(b)(2) of the Immigration and Nationality Act (“the Act”), with a request for a National Interest Waiver of the job offer and labor certification requirements under section 212(a)(5)(A) of the Act.

Note on Supporting Evidence: This petition includes a Letter of Recommendation from Corey Graham, Executive Director of Software Engineering at JPMorgan Chase & Co. (Exhibit WP-1). This letter was originally drafted to attest to Mr. Vivek Mishra’s Extraordinary Ability for an EB-1A petition. *The current petition is filed under the EB-2 National Interest Waiver standard*, which requires demonstration that (1) the proposed endeavor has substantial merit and national importance, (2) the beneficiary is well positioned to advance that endeavor, and (3) on balance, it would benefit the United States to waive the job offer requirement. Mr. Graham’s letter is submitted as evidence supporting Prongs 1 and 2 of the *Dhanasar* analysis—specifically, that Mr. Vivek Mishra’s work has substantial merit (mission-critical SIFI infrastructure) and that he has a proven track record of success in high-stakes environments, positioning him well to advance the proposed endeavor.

This petition demonstrates that Mr. Vivek Mishra meets the requirements for the National Interest Waiver (NIW) as established by the Administrative Appeals Office in *Matter of Dhanasar*, 26 I&N Dec. 884 (AAO 2016). Specifically, this petition provides evidence that:

1. Mr. Vivek Mishra possesses an advanced degree (Master of Science in Computer Science from California State University, Long Beach), satisfying the threshold requirement for EB-2 classification under 8 C.F.R. § 204.5(k)(2).
2. Mr. Vivek Mishra satisfies the three-prong test established in *Matter of Dhanasar*:

- **Prong 1:** Mr. Vivek Mishra’s proposed endeavor has both substantial merit and national importance. (Section 4)
- **Prong 2:** Mr. Vivek Mishra is well positioned to advance the proposed endeavor. (Section 5)
- **Prong 3:** On balance, it would be beneficial to the United States to waive the requirements of a job offer and labor certification. (Section 6)

Pursuant to 8 C.F.R. § 204.5(k), Mr. Vivek Mishra may file an I-140 visa petition for classification under Section 203(b)(2) of the Act as an alien with an advanced degree seeking a National Interest Waiver. Pursuant to 8 C.F.R. § 204.5(k)(4)(ii), Mr. Vivek Mishra requests that the requirement of a job offer and labor certification be waived because his admission to the United States would be in the national interest.

1 Executive Summary

This petition demonstrates that Mr. Vivek Mishra satisfies all requirements for the National Interest Waiver under the three-prong *Dhanasar* framework:

1.1 Prong 1: Substantial Merit and National Importance

Mr. Vivek Mishra’s proposed endeavor—applying AI-driven intelligent systems to enhance the resilience and optimization of U.S. critical infrastructure—has both substantial merit and national importance:

- **Enterprise-Scale Production Systems:** Architected high-availability distributed infrastructure at JPMorgan Chase (SIFI) processing over 5 million consumer requests monthly with 99.9% reliability, protecting economic stability for 80+ million Americans. (Exhibits NI-1, NI-2)
- **Federally Funded Research:** Conducted U.S. DOT-funded intelligent transportation systems research, with two publications permanently indexed in the National Transportation Library (ROSA P Records dot/61210, dot/59030). (Exhibits NI-3, NI-4, SR-4, SR-8)
- **Cross-State Research Adoption:** Methodologies adopted by Florida DOT-funded research, demonstrating real-world applicability to state infrastructure planning. (Exhibit WP-2)
- **Knowledge Dissemination:** Authored technical book on AI agent design (ISBN: 979-8268912371) adopted by accredited academic institution for curriculum use. (Exhibits NI-7, NI-8)
- **National Policy Alignment:** Work directly supports Justice40 Initiative (transportation equity for underserved communities) and Executive Order 14110 on safe, trustworthy AI deployment. (Exhibits NI-3, NI-5)
- **SIFI Infrastructure Protection:** Contributions to JPMorgan Chase systems critical to U.S. financial stability under Dodd-Frank designation. (Exhibit NI-6)

1.2 Prong 2: Well Positioned to Advance the Endeavor

Mr. Vivek Mishra is well positioned to advance this endeavor based on education, demonstrated success, and ongoing progress:

- **Advanced Degree:** MS in Computer Science from California State University, Long Beach (R2 Doctoral University—High Research Activity). (Exhibit AD-1)
- **Production Track Record:** Currently deployed systems serving 80+ million users with documented 99.9% reliability; recruited by Civil Engineering department specifically for AI/Python expertise to bridge academic research with infrastructure applications. (Exhibits NI-2, WP-8)
- **Research Impact:** 9 peer-reviewed publications with 37 citations; permanent federal indexing in National Transportation Library; methodologies adopted by state DOT-funded projects; most recent publication on AI-driven cloud infrastructure optimization for financial systems (IJARSCT, Impact Factor: 8.2, February 2026). (Exhibits SR-1 through SR-9, WP-9)
- **Professional Recognition:** IEEE Senior Member (top 8% of 460,000 members), AWS Certified Cloud Practitioner, peer reviewer for premier ML conferences (ICLR 2025, NeurIPS 2025). (Exhibits WP-5, WP-6, ER-1, ER-2)
- **Expert Validation:** Independent attestation from Executive Director at JPMorgan Chase (23 years experience), Associate Professor with Florida DOT-funded research, and CSULB faculty regarding specialized qualifications and demonstrated impact. (Exhibits WP-1, WP-2, WP-3)

1.3 Prong 3: On Balance, Beneficial to Waive Requirements

On balance, waiving the job offer and labor certification requirements would benefit the United States:

- **Cross-Domain Flexibility:** Endeavor spans financial infrastructure and transportation systems, requiring adaptability across multiple projects that a single PERM job description cannot accommodate.
- **Ongoing Critical Systems:** Production systems protecting 80+ million consumers would face disruption during 12-18 month PERM process, impacting SIFI stability.
- **Specialized Expertise:** Combination of AI agent design, distributed systems engineering, and infrastructure optimization represents rare skill set in current AI/ML labor market.
- **Research Continuity:** Ongoing collaborations across transportation and financial domains benefit from flexible engagement not constrained by narrow job descriptions.
- **No Adverse Impact:** Specialized nature of work and documented AI/ML talent constraints indicate no displacement of U.S. workers.

2 The Proposed Endeavor

Mr. Vivek Mishra proposes to continue advancing the development and deployment of AI-driven intelligent systems to enhance the resilience and optimization of U.S. critical infrastructure. His work applies agent-based modeling, distributed systems engineering, and machine learning to protect essential national systems—currently focused on financial infrastructure at JPMorgan Chase and continuing research in intelligent transportation systems.

2.1 Unified Endeavor Across Infrastructure Domains

This endeavor represents a coherent application of AI and intelligent-systems methodologies to two critical infrastructure domains:

Financial Infrastructure (Current Primary Focus): At JPMorgan Chase & Co.—a Systemically Important Financial Institution (SIFI) under Dodd-Frank—Mr. Vivek Mishra architects and operates high-availability distributed systems that process over 5 million consumer requests monthly with 99.9% reliability. These systems protect economic stability for 80+ million American consumers through secure transaction processing, architected fraud prevention service integration, and reliable vendor integrations. The infrastructure directly supports U.S. financial system stability as designated critical by the Financial Stability Oversight Council. (Exhibits NI-1, NI-2, NI-6)

Transportation Infrastructure (Research and Knowledge Dissemination): Mr. Vivek Mishra’s federally funded research on intelligent transportation systems—permanently indexed in the National Transportation Library (ROSA P Records dot/61210, dot/59030)—developed mathematical frameworks for applying machine learning to infrastructure optimization. This work, subsequently adopted by Florida DOT-funded research, supports Justice40 Initiative goals for transportation equity and provides decision-support tools for public infrastructure investments. (Exhibits NI-3, NI-4, SR-4, SR-8, WP-2)

Technical Bridge: Both domains employ the same core methodologies: designing resilient distributed systems under reliability and security constraints, applying machine learning for optimization and performance analysis, and implementing agent-based approaches to complex infrastructure challenges. Mr. Vivek Mishra’s recruitment by CSULB’s Civil Engineering Department specifically for AI and Python expertise (Exhibit WP-8) demonstrates the cross-disciplinary applicability of these methods.

2.2 National Importance and Policy Alignment

This endeavor advances critical national priorities:

- **Executive Order 14110:** Demonstrates practical application of AI safety principles to protect financial stability and infrastructure resilience, directly supporting the Administration’s emphasis on safe, secure, and trustworthy AI deployment.
- **SIFI Protection:** Contributes to the stability of U.S. financial infrastructure through systems protecting millions of consumers from fraud and ensuring reliable access to financial resources.
- **Justice40 Initiative:** Transportation research directly supports federal commitment to deliver 40% of infrastructure benefits to disadvantaged communities through accessibility optimization.

- **Knowledge Infrastructure:** Federal archiving of research and book adoption by academic institutions supports broader AI safety and infrastructure resilience knowledge base. (Exhibits NI-7, NI-8)

3 Mr. Vivek Mishra Possesses an Advanced Degree

Pursuant to 8 C.F.R. § 204.5(k)(2), Mr. Vivek Mishra qualifies for EB-2 classification based on his possession of an advanced degree.

Educational Credentials:

- **Master of Science in Computer Science** (2021)
California State University, Long Beach
Carnegie Classification: R2 Doctoral University (High Research Activity)
Evidence: Exhibit AD-1
- **Bachelor of Technology in Information Technology** (2014)
West Bengal University of Technology, India
Evidence: Exhibit AD-2

Mr. Vivek Mishra’s Master’s degree in Computer Science from California State University, Long Beach satisfies the advanced degree requirement under 8 C.F.R. § 204.5(k)(2), which defines an advanced degree as “any United States academic or professional degree or a foreign equivalent degree above that of baccalaureate.”

California State University, Long Beach is classified as an R2 Doctoral University under the Carnegie Classification system, indicating “High Research Activity.” This classification places CSULB among the top tier of U.S. research universities.

Conclusion: Mr. Vivek Mishra clearly satisfies the advanced degree requirement for EB-2 classification.

4 Prong 1: The Proposed Endeavor Has Substantial Merit and National Importance

Under the first prong of the *Dhanasar* framework, Mr. Vivek Mishra must demonstrate that his proposed endeavor has both substantial merit and national importance.

4.1 Substantial Merit of the Proposed Endeavor

4.1.1 Enterprise-Scale Production Systems

Mr. Vivek Mishra architects and operates high-availability distributed infrastructure at JPMorgan Chase processing over 5 million consumer requests monthly with 99.9% reliability, serving 80+ million Americans. (Exhibits NI-1, NI-2)

Executive Director Corey Graham (23 years experience at JPMC) attests:

“Throughout the development of the Critical Financial Infrastructure Platform, Vivek played a critical role in building and deploying the core backend services that underpin the platform. He designed APIs that are now part of a mission-critical system, implemented observability and alerting infrastructure to ensure uptime and reliability, and managed secure integrations with external vendors for rewards redemption and settlement.” (Exhibit WP-1)

4.1.2 Federally Funded Research

Mr. Vivek Mishra conducted U.S. DOT-funded research on intelligent transportation systems and published AI-driven optimization frameworks for financial infrastructure, resulting in 9 peer-reviewed publications with 37 citations from researchers across the United States, Italy, Iran, Romania, and other nations. Two publications are permanently indexed in the National Transportation Library (ROSA P Records dot/61210, dot/59030). His most recent publication, “AI-Driven Optimization of Backend and Cloud Infrastructure in Large-Scale Financial Systems” (IJARSCT, Impact Factor: 8.2, February 2026), directly bridges his enterprise experience at JPMorgan Chase with rigorous academic frameworks for reinforcement learning-based autoscaling, learned data layer optimization, and AI-driven incident prediction in regulated financial environments. (Exhibits NI-3, NI-4, SR-1 through SR-9)

Dr. Maxim A. Dulebenets, Associate Professor at FAMU-FSU College of Engineering with Florida DOT-funded research, confirms:

“Vivek’s work has shaped aspects of our freight network modeling and equity-focused analyses in Florida, particularly in how we evaluate accessibility and performance under different policy constraints. [...] Vivek brings an interdisciplinary depth, combining AI, programming, and infrastructure modeling, that is both rare and increasingly essential in modern transportation research.” (Exhibit WP-2)

4.1.3 Knowledge Dissemination

Mr. Vivek Mishra authored “Design Patterns for AI Agents: A Structured Approach to Building Robust and Intelligent Systems” (ISBN: 979-8268912371), an early structured engineering framework for AI agent architecture, adopted by an accredited academic institution for curriculum use. (Exhibits NI-7, NI-8)

4.2 National Importance of the Proposed Endeavor

4.2.1 SIFI Infrastructure Protection

JPMorgan Chase & Co. is designated as a Systemically Important Financial Institution (SIFI) by the Financial Stability Oversight Council under Dodd-Frank, with \$4.1 trillion in assets, 84 million consumer accounts, and 6 million small business accounts. Mr. Vivek Mishra’s infrastructure work directly protects U.S. financial system stability:

- **Fraud Prevention:** Architected and implemented integration of fraud prevention services, protecting consumer accounts from fraudulent activity at scale
- **System Reliability:** 99.9% uptime ensures continuous consumer access to financial resources
- **Cascading Risk Mitigation:** Prevents disruptions that could trigger instability throughout national and global financial systems

(Exhibits NI-6, NI-1, NI-2)

4.2.2 Federal Transportation Knowledge Infrastructure

Permanent indexing in the National Transportation Library (official DOT repository operated by Bureau of Transportation Statistics) recognizes Mr. Vivek Mishra’s research as a contribution to national transportation knowledge infrastructure supporting billions of dollars in federal investments. (Exhibits NI-3, NI-4)

4.2.3 Justice40 Initiative Alignment

Research on optimizing multimodal transportation access for low-income transit riders directly supports the Administration’s Justice40 Initiative commitment to deliver 40% of federal infrastructure benefits to disadvantaged communities. Methodologies have been adopted by state DOT-funded projects. (Exhibits NI-5, WP-2)

4.2.4 Executive Order 14110 on Safe AI

Work developing production AI systems, authoring technical frameworks for AI agent design, and conducting peer-reviewed AI research directly advances Executive Order 14110 priorities for safe, secure, and trustworthy AI deployment critical to national security and economic competitiveness.

4.3 Conclusion: Prong 1 Satisfied

Mr. Vivek Mishra’s proposed endeavor has both substantial merit (enterprise-scale AI systems, peer-reviewed research, technical book authorship) and national importance (SIFI financial infrastructure, federal transportation priorities, AI national competitiveness). The first prong of the *Dhanasar* test is satisfied.

5 Prong 2: Mr. Vivek Mishra is Well Positioned to Advance the Proposed Endeavor

Under the second prong of the *Dhanasar* framework, Mr. Vivek Mishra must demonstrate that he is well positioned to advance the proposed endeavor.

5.1 Education and Credentials

Mr. Vivek Mishra possesses qualifications directly relevant to advancing AI-driven infrastructure protection:

- MS in Computer Science from California State University, Long Beach (R2 Doctoral University—High Research Activity)
- AWS Certified Cloud Practitioner for production cloud systems
- IEEE Senior Member (top 8% of 460,000 members), demonstrating significant performance over 5+ years

(Exhibits AD-1, WP-5, WP-6)

5.2 Production Track Record

At JPMorgan Chase, Mr. Vivek Mishra has successfully deployed AI-integrated systems to protect critical financial infrastructure, positioning him well to continue this work:

- Architected high-availability distributed infrastructure serving 80+ million Americans (deployed June 2025)
- Built systems processing 5+ million consumer requests monthly with 99.9% reliability
- Implemented observability infrastructure for mission-critical SIFI systems
- Architected and implemented fraud prevention service integration into distributed infrastructure, ensuring transaction integrity and consumer protection at scale

Director Corey Graham attests: “*What sets Vivek apart is his ability to approach complex engineering challenges with both precision and foresight. [...] His work has already been cited in academic research and has directly contributed to the development of nationally deployed platforms at one of the largest banks in the United States.*” (Exhibit WP-1)

5.3 Research Impact

Mr. Vivek Mishra’s research demonstrates the academic foundation positioning him to advance infrastructure optimization:

- 9 peer-reviewed publications with 37 citations from international researchers
- Two publications permanently indexed in National Transportation Library

- Methodologies adopted by Florida DOT-funded research
- U.S. DOT-funded work on California High-Speed Rail (\$105 billion infrastructure project)

Dr. Maxim Dulebenets (Associate Professor, FAMU-FSU, Florida DOT-funded research): “*Vivek brings an interdisciplinary depth, combining AI, programming, and infrastructure modeling, that is both rare and increasingly essential in modern transportation research.*” (Exhibit WP-2)

5.4 Thought Leadership

Mr. Vivek Mishra has established himself as a knowledge contributor, positioning him to disseminate advances in AI safety and infrastructure protection:

- Authored technical book on AI agent design adopted by accredited academic institution
- Peer reviewer for premier ML conferences (ICLR 2025, NeurIPS 2025)
- Featured as IEEE Impact Creator on thought leadership platform
- Invited Eminent Speaker at AICTE-approved Faculty Development Programme

(Exhibits NI-7, NI-8, ER-1, ER-2, WP-7)

5.5 Ongoing Progress

Mr. Vivek Mishra has already made substantial progress toward the proposed endeavor:

- ✓ Currently developing AI-integrated platforms at JPMorgan Chase (SIFI)
- ✓ Systems serving millions with documented reliability metrics
- ✓ Research indexed in federal repositories and adopted by state agencies
- ✓ Technical book published and adopted by academic institution

5.6 Conclusion: Prong 2 Satisfied

Mr. Vivek Mishra’s education, production track record at a SIFI, federally recognized research, thought leadership roles, and ongoing progress collectively demonstrate that he is well positioned to advance the proposed endeavor of applying AI-driven intelligent systems to U.S. critical infrastructure protection.

6 Prong 3: On Balance, It Would Be Beneficial to the United States to Waive the Job Offer and Labor Certification Requirements

Under the third prong of the *Dhanasar* framework, USCIS must determine whether, on balance, it would be beneficial to the United States to waive the job offer and labor certification requirements.

6.1 Cross-Domain Nature of the Endeavor

Mr. Vivek Mishra’s proposed endeavor spans financial infrastructure and transportation systems, requiring flexibility that a single PERM job description cannot accommodate:

- Work applies the same AI and intelligent-systems methodologies across multiple infrastructure domains
- Financial infrastructure projects (SIFI protection) and transportation research (federally funded) benefit from unified technical approach
- Peer-reviewed research collaborations across universities and government agencies require engagement not constrained by narrow employer-specific job descriptions
- AI agent design and distributed systems work evolves across projects as technology advances

The labor certification process, which typically requires 12-18 months and ties workers to specific job descriptions, would introduce rigidity incompatible with the cross-disciplinary, evolving nature of AI infrastructure work.

6.2 Rare Combination of Specialized Expertise

Mr. Vivek Mishra possesses a combination of qualifications representing a rare skill set in the current AI/ML labor market:

- Production engineer at SIFI with documented impact serving 80+ million consumers
- Peer-reviewed researcher with federal indexing (National Transportation Library)
- IEEE Senior Member and peer reviewer for premier ML conferences (ICLR, NeurIPS)
- Technical author with academic institution adoption of AI safety frameworks

This integration of research credentials (federal recognition, peer review), production experience (Fortune 500 SIFI), and thought leadership (IEEE, academic adoption) positions Mr. Vivek Mishra to make contributions that benefit the United States in ways extending beyond typical employment arrangements.

6.3 Ongoing Critical Systems Would Face Disruption

Mr. Vivek Mishra currently develops and maintains production systems protecting 80+ million consumers:

- Systems process 5+ million requests monthly with 99.9% reliability

- Infrastructure serves a SIFI designated critical to U.S. financial stability
- Architected fraud prevention service integration protecting consumer economic well-being at scale

A 12-18 month PERM process would introduce uncertainty during active development of infrastructure supporting tens of millions of Americans. On balance, allowing continuation of this work without delay better serves the national interest in financial system resilience.

6.4 Benefits to U.S. Interests

Waiving the job offer and labor certification requirements would benefit the United States by:

- **Infrastructure Continuity:** Enables uninterrupted contribution to SIFI infrastructure protecting 80+ million consumers
- **Research Flexibility:** Allows continued collaboration on federally funded transportation research and cross-domain knowledge transfer
- **Knowledge Dissemination:** Facilitates thought leadership through academic institution adoption, IEEE platforms, and peer review service
- **Policy Alignment:** Supports Executive Order 14110 and Justice40 Initiative goals without employer-locked constraints

6.5 No Adverse Impact on U.S. Workers

The specialized nature of Mr. Vivek Mishra's work and documented AI/ML talent constraints indicate no displacement of U.S. workers:

- Combination of SIFI production engineering, federally recognized research, and AI safety thought leadership represents rare specialization
- Contributions to peer-reviewed research, academic curriculum development, and industry knowledge sharing benefit the broader AI/ML workforce
- Work addresses national-level infrastructure challenges (SIFI protection, federal transportation systems) rather than routine commercial development

6.6 Conclusion: Prong 3 Satisfied

On balance, the national interest in Mr. Vivek Mishra's continued contributions to critical infrastructure protection—spanning financial system stability (SIFI), federally funded transportation research, and AI safety advancement—outweighs the protections afforded by the labor certification process. The cross-domain nature of the endeavor, rare combination of specialized expertise, ongoing critical systems serving tens of millions of Americans, and absence of adverse impact on U.S. workers collectively support waiving the job offer and labor certification requirements.

7 Conclusion: National Interest Waiver Should Be Granted

This petition has demonstrated that Mr. Vivek Mishra satisfies all requirements for EB-2 classification with a National Interest Waiver:

7.1 EB-2 Qualification

- ✓ **Advanced Degree:** MS in Computer Science from California State University, Long Beach (R2 Doctoral University)

7.2 Dhanasar Three-Prong Analysis

- ✓ **Prong 1 - Substantial Merit and National Importance:**

- Enterprise AI systems serving 84 million Americans through SIFI financial institution
- Research permanently indexed in National Transportation Library
- Alignment with Justice40 Initiative and Executive Order 14110 on AI
- Technical book providing an early structured framework for AI agent architecture

- ✓ **Prong 2 - Well Positioned to Advance:**

- MS Computer Science from R2 Doctoral University
- IEEE Senior Member (top 8% of 460,000 members)
- Current position at JPMorgan Chase developing AI systems
- 9 peer-reviewed publications with federal funding and recognition
- Record of success with documented production metrics

- ✓ **Prong 3 - Beneficial to Waive Requirements:**

- Critical financial infrastructure cannot await labor certification
- Unique combination of research and industry credentials
- Labor certification would impede flexible AI development
- No adverse impact on U.S. workers

7.3 Final Determination

Based on the evidence presented, Mr. Vivek Mishra has demonstrated that:

1. He possesses an advanced degree qualifying him for EB-2 classification;
2. His proposed endeavor in AI/ML for financial technology and transportation infrastructure has substantial merit and national importance;
3. He is well positioned to advance the proposed endeavor based on his education, skills, record of success, and ongoing progress; and

4. On balance, it would be beneficial to the United States to waive the job offer and labor certification requirements.

Accordingly, this petition respectfully requests that USCIS approve Mr. Vivek Mishra's I-140 petition for EB-2 classification with a National Interest Waiver.

Respectfully submitted,

Vivek Kumar Mishra
Self-Petitioner

List of Exhibits

Advanced Degree Evidence

- Exhibit 1: **Exhibit AD-1:** Master of Science Degree in Computer Science.
Official diploma and transcripts from California State University, Long Beach (2021).
- Exhibit 2: **Exhibit AD-2:** Bachelor of Technology Degree in Information Technology.
Official diploma from West Bengal University of Technology (2014).
- Exhibit 3: **Exhibit AD-3:** Credential Evaluation (if applicable).
Third-party credential evaluation confirming U.S. equivalency.

Evidence of Substantial Merit and National Importance (Prong 1)

- Exhibit 4: **Exhibit NI-1:** JPMorgan Chase Press Release - Critical Financial Infrastructure Platform Launch.
Official press release announcing the deployment of high-availability distributed infrastructure serving 80+ million Chase cardmembers (June 2025).
- Exhibit 5: **Exhibit NI-2:** Critical Financial Infrastructure Production Metrics (Splunk Dashboard).
Screenshot from Splunk Enterprise showing production metrics: 5,012,443 user requests, 22,623 orders, and 99.9% success rate.
- Exhibit 6: **Exhibit NI-3:** National Transportation Library ROSA P Record dot/61210.
Federal repository record for “Optimizing Multimodal Transportation Access to Support Commuting Among Low-Income Transit Riders.”
- Exhibit 7: **Exhibit NI-4:** National Transportation Library ROSA P Record dot/59030.
Federal repository record for “Evaluating Innovative Financing Mechanisms for the California High-Speed Rail Project.”
- Exhibit 8: **Exhibit NI-5:** Executive Order 14110 on Safe, Secure, and Trustworthy AI.
White House Executive Order establishing AI as national priority (October 30, 2023).
- Exhibit 9: **Exhibit NI-6:** JPMorgan Chase SIFI Designation Documentation.
Evidence of JPMorgan Chase’s designation as Systemically Important Financial Institution under Dodd-Frank.
- Exhibit 10: **Exhibit NI-7:** Book Adoption Letter from Anand International College of Engineering.
Official letter confirming adoption of Mr. Vivek Mishra’s AI Agents book into permanent library collection.
- Exhibit 11: **Exhibit NI-8:** Design Patterns for AI Agents - Technical Book.
Full text of “Design Patterns for AI Agents: A Structured Approach to Building Robust and Intelligent Systems” (ISBN: 979-8268912371).

Evidence of Being Well Positioned to Advance (Prong 2)

- Exhibit 12: **Exhibit WP-1:** Letter of Recommendation from Corey Graham.
Detailed letter from Director of Software Engineering at JPMorgan Chase & Co. (23 years experience) attesting to Mr. Vivek Mishra's critical contributions.
- Exhibit 13: **Exhibit WP-2:** Letter of Recommendation from Dr. Maxim A. Dulebenets.
Expert testimonial from Associate Professor at FAMU-FSU College of Engineering regarding Mr. Vivek Mishra's research impact.
- Exhibit 14: **Exhibit WP-3:** Letter of Recommendation from Dr. Shailesh Chandra.
Expert testimonial from Associate Professor at California State University, Long Beach.
- Exhibit 15: **Exhibit WP-4:** Letter of Recommendation from Dr. Zhenyu Wang.
Expert testimonial from Senior Research Associate at University of South Florida.
- Exhibit 16: **Exhibit WP-5:** IEEE Senior Member Certification.
Official IEEE Senior Member certificate and membership card (Member Number: 100495418).
- Exhibit 17: **Exhibit WP-6:** AWS Certified Cloud Practitioner Certificate.
Professional certification demonstrating cloud technology expertise.
- Exhibit 18: **Exhibit WP-7:** Letter of Recommendation from Rahul Vishwakarma.
Detailed letter from Research Engineer at WorkOnward (Techstars-backed), holder of 60+ U.S. patents in AI, data processing, and distributed systems.
- Exhibit 19: **Exhibit WP-8:** Research Assistant Offer Letter from California State University, Long Beach.
Official offer letter from Dr. Shailesh Chandra, Assistant Professor at CSULB, documenting research assistantship position (February 2020).
- Exhibit 20: **Exhibit WP-9:** Google Scholar Citation Profile.
Screenshot documenting 37 total citations, H-index of 3, and publication record.
- Exhibit 21: **Exhibit WP-10:** Curriculum Vitae.
Complete CV documenting education, experience, publications, and achievements.

Scholarly Publications

- Exhibit 22: **Exhibit SR-1:** Integrating Solar Photovoltaic Systems into the Grid: An Overview of AI Application.
Peer-reviewed scholarly article on AI applications for renewable energy.
- Exhibit 23: **Exhibit SR-2:** The Role of Generative AI in Revolutionizing Healthcare, Education, and Finance.
Peer-reviewed scholarly article on generative AI applications.

- Exhibit 24: **Exhibit SR-3:** MDPI Sustainability Journal Publication (sustainability-16-08077).
Peer-reviewed article published in MDPI Sustainability (Impact Factor: 3.9).
- Exhibit 25: **Exhibit SR-4:** Framework for Rail Transport Inequality Assessment.
Most cited publication with 15 citations.
- Exhibit 26: **Exhibit SR-5:** Evaluating Financing Mechanisms and Economic Benefits to Fund Grade Separation Projects.
Peer-reviewed publication with 12 citations.
- Exhibit 27: **Exhibit SR-6:** Evaluating Innovative Financing Mechanisms for the California High-Speed Rail Project.
Peer-reviewed scholarly article with 7 citations.
- Exhibit 28: **Exhibit SR-7:** Funding the High-Speed Rail: A Case Study of the California Project.
Peer-reviewed publication with 3 citations.
- Exhibit 29: **Exhibit SR-8:** Optimizing Multimodal Transportation Access to Support Commuting Among Low-Income Transit Riders.
Peer-reviewed article on transportation equity.
- Exhibit 30: **Exhibit SR-9:** AI-Driven Optimization of Backend and Cloud Infrastructure in Large-Scale Financial Systems.
Peer-reviewed scholarly article published in IJARST (Impact Factor: 8.2), Volume 6, Issue 2, February 2026 (DOI: 10.48175/IJARST-31286).

Evidence of Expert Recognition

- Exhibit 31: **Exhibit ER-1:** ICLR 2025 FM-Wild Workshop Reviewer Invitation and Completion.
Evidence of peer review service at top-tier ML conference.
- Exhibit 32: **Exhibit ER-2:** NeurIPS 2025 Position Paper Track Reviewer Evidence.
Evidence of peer review service at premier ML conference.
- Exhibit 33: **Exhibit ER-3:** IEEE Senior Member Application Review Panel Service.
Evidence of service on IEEE membership evaluation panel (June 2025).
- Exhibit 34: **Exhibit ER-4:** IEEE Transmitter Impact Creators Profile.
Official IEEE thought leadership platform profile.
- Exhibit 35: **Exhibit ER-5:** IEEE.tv Speaker Profile Page.
Official speaker profile on IEEE's television network.
- Exhibit 36: **Exhibit ER-6:** Faculty Development Programme Speaker Schedule.
Evidence of "Eminent Speaker" invitation at AICTE-approved institution.

Exhibit AD-1

Master of Science Degree in Computer Science

[MS Diploma and Transcripts from CSULB]

California State University, Long Beach (2021)

Carnegie Classification: R2 Doctoral University

Exhibit AD-2

Bachelor of Technology Degree in Information Technology

[B.Tech Diploma from WBUT]
West Bengal University of Technology (2014)

Exhibit NI-1

JPMorgan Chase Press Release - Critical Financial Infrastructure Platform
Launch